Find the Promos

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Why Problem 1

The brief names the bottleneck: an editor watches every episode and picks promo moments by hand, against a weekly release cadence — a recurring cost with a checkable output. I scored all three problems against the five lenses plus feasibility and measurability (`docs/02_decision.md`); only valuing raw feasibility above everything flipped the winner. The risk I accepted is that it's the obvious pick, so I differentiated on method rather than topic.

Assumptions where the brief was silent

A promo is a **cut list**, not a rendered clip. 20–60s and "hook inside 3 seconds" are **hard constraints in code**. Timestamps are **relative to each episode's start** (verified: all 50 begin at `00:00.00`). Splicing is allowed within an episode, but only **one** promo is built across episodes — stitching across 50 needs character identity this corpus can't support. Episode runtimes contradict the "1–2 minute" framing (median 118.9s, **12 of 50 over 150s**, max 200.8s); I treated the data as truth. There is **no ground truth** for "good promo", so weights are hand-set and documented, never described as learned.

How it works

Code owns time; the model owns judgment. The model never sees a timestamp and never writes one — it reads beats as `[b07] JULIANA (desperate): "..."` and answers in beat IDs, and code looks each ID up in the parsed table. That makes a hallucinated timestamp impossible rather than merely unlikely.

Parse *(code)* — 50 files → **3,727 beats**. Reconciles labels vs timestamps vs beats and aborts on mismatch; reports **22 source hazards**; merges **64 speaker labels → 58 characters**, stripping an honorific only when the bare name is attested elsewhere, so `LORD` and `LADY ASHCROFT` stay two people.

Score *(code — the unfair advantage)* — the scripts already label delivery, `(Cold, Commanding)`, `(Screaming, Desperate)`, on **1,472 of 1,487 dialogue beats** across 617 values, so the tool never guesses where the drama is. A hand-written lexicon (90% hit rate) builds an intensity curve and a promo is a shape on it: **18,141** legal windows on real beat boundaries → **3,376** rejected for no hook in the first 3s → **171** shortlisted. 0.8s, offline, free.

Select *(model, 1 call/episode)* — pick among pre-validated windows; write hook, cliffhanger, one-line why, and a hook type from DramaBox's taxonomy (moral outrage / romantic tension / empathy shock / power reversal) so an editor can spread types instead of posting four of a kind.

Splice *(code)* — two-piece promos within one episode, 23–77s of middle removed, rejected unless the pair has an arc: the speaker changes or the intensity rises. **41 of 91** candidates were two-piece.

Validate *(code)* — 6 hard checks, plus duplicate suppression on **visual** text between episodes 2 apart: 37 dropped, of which 10 are genuine cross-episode repeats including episodes 31 and 32, which both close on a branding iron inches from the heroine's eye.

Gate *(model, 1 call/promo)* — a fresh instance sees **only the clip**, no logline or episode number or surrounding beats, and must say who this is, what's at stake, what question remains, and whether it would keep watching.

The tool never posts: it turns "watch 50 episodes and scrub" into "review 47 ranked, timestamped candidates", and the editor's keep/reject log is the dataset that would let the weights stop being my opinion.

Scale and cost

Measured: one 50-episode show = **127 calls, 124k in + 75k out tokens**; at a show a week (2,600 episodes/yr) ~6,600 calls and ~10.4M tokens. **$0 on free tier**, ~**$2.20/yr** at Gemini Flash-Lite rates. **Tokens are not the scaling cost** — the binding constraint is the free tier's 8,000 tokens/minute, which paces a show to ~20 min of wall clock and vanishes on a paid key. The real cost is editor attention, which is why the gate and duplicate suppression exist at all.

How I know it's good

Validators, over all 92 shipped + rejected + flagship promos: `timestamps_exist` **92/92** — zero invented timestamps — and **92/92** on each of the other five. **Blind cold-viewer gate**: 54 tested, **7 failed**, pass rate **0.870**, none untested; per criterion, unresolved-ending 1.00, character-identifiable 0.91, stakes-clear 0.76, **would-keep-watching 0.65**. **Determinism**: two runs produce the identical decision fingerprint `c7b809ad92b976e7`. **Coverage**: 42 of 50 episodes ship at least one promo.

Does it beat picking at random? This is the measure I'd defend, because it needs no human time and runs on any new show. Each pick is compared against a **random legal window from the same episode** — same length rules, same real beat boundaries, just not selected. A blind judge sees two clips as text, never a score, never which is which.

Compared against	Position-corrected win rate	Counterbalanced subset
Any legal 20–60s window — tests the whole system	**75.8%**	**78.6%** (tool won both ways on 10 of 14 pairs)
A window that **already passes the hook filter** — tests what the tension scoring adds	**65.1%**	**64.3%**

Chance is 50%, n=38 pairs, uncorrected p=0.0017. The judge turned out to strongly prefer whichever clip came **second** (it won 88.5% there versus 41.7% shown first), so I re-judged as many pairs as the token budget allowed in the opposite order. Two independent corrections — averaging the position-conditional rates, and a strictly counterbalanced subset — agree. **The honest decomposition: requiring a strong opening does most of the work, and the tension scoring plus model selection adds a real but smaller edge on top.** Only 2 of 14 pairs flipped when order was swapped.

Does splicing cost comprehension? The brief says a promo is *usually* one continuous stretch; the gate shows why. Continuous promos pass at **0.886** (mean score 3.36), spliced at **0.800** (3.10). I had charged a 0.10 standalone penalty for the cut on instinct — this measurement is what justifies it, and no spliced promo reaches the top of the ranking.

Why there is no hand-picked gold set. I built one, then demoted it. A reference set asks "did the tool match one human's pick?", which is not the job, and maintaining one per show would recreate the manual work the tool removes. With n=6 the per-episode spread was larger than the differences between approaches, so it could not rank them. It survives as a sanity check only (`docs/gold_set.json`, mean IoU 0.465). **How this validates once a real team uses it:** the editor's keep/reject decisions in the UI are logged, which is both the acceptance-rate metric and the dataset that would replace my hand-set weights with real preferences — at no extra cost, because reviewing the shortlist is already the job.

Cross-model stability (`gpt-oss-120b` vs `qwen3.8-27b`, 50 episodes each): mean IoU **0.50**, identical picks 17/50, **no overlap at all on 8/50**, hook-type agreement **0.48** on a four-way label. A real part of this ranking is model taste, not signal — keep the editor in the loop and treat cross-model disagreement as a confidence flag. `qwen` also produced 1 validator failure where `gpt-oss-120b` produced none.

Where it fails

1. It cannot see — shot quality, performance and music are invisible in text; I'm reading sheet music and guessing at the performance. **2.** The evaluation is too small to answer its own main question (n=6). **3.** "Would keep watching" is the weakest criterion at 0.65: comprehensible but not always compelling, and that gap is taste. **4.** 8 of 50 episodes ship nothing, losing their best candidate to the gate or to duplicate suppression. **5.** The duplicate check nearly shipped dead — my first threshold (0.45 on dialogue text) could never fire, because across 1,225 pairs dialogue overlap peaks at 0.177: repetition here is visual, not verbal. **6.** The alias map is partly judgment — `ALEXANDER'S MOTHER` / `DOWAGER DUCHESS` / `MATRIARCH` are plausibly one woman, left unmerged for want of evidence.

AI tools used, and where they got it wrong

Claude for code and drafting; Groq-hosted `openai/gpt-oss-120b` in the pipeline, `qwen/qwen3.8-27b` as comparison. Full log: `docs/ai_mistakes_log.md`.

The one that matters. On episode 8, `qwen3.8-27b` chose a promo spanning `00:25.00 → 00:49.60` and named as its hook a beat starting **8.7 seconds before the promo begins**. The hook text was fluent and that moment genuinely exists in the episode — it just isn't in the clip. No reader could catch that; only checking the beat table can, and two validators did. The same model, on a three-beat test, returned `start_beat: b_03, end_beat: b_01` — a promo that plays backwards, in valid JSON.

The one that nearly fooled me, in my own code. When splicing landed, gold agreement jumped 0.46 → 0.519, looking like a win. My IoU used a promo's outer envelope, crediting a two-piece promo with covering the seconds the splice *removes*. Fixed to measure the real segment union; agreement returned to 0.465. Nothing errored. A metric that improves when you add code deserves more suspicion than one that doesn't.